

Amanda Achiangia

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EDUCATION

York University — B.Sc. Specialist Honours, Applied Mathematics (Financial Mathematics) Expected June 2026

GPA: 3.3/4.0 (B+) | Coursework: Models for Financial Economics, Mathematical Theory of Interest, Stochastic Processes, Risk Management & Insurance, Regression Analysis (A+), Numerical Methods I & II, Advanced Numerical Methods, Differential Equations, Real Analysis, Corporate Finance, Scientific Computation for Finance

TECHNICAL SKILLS

Capital Markets	Derivatives pricing, fixed income analysis, FX, equity analysis, Monte Carlo simulation, VaR, Expected Shortfall, statistical backtesting (Kupiec POF, Christoffersen), stress testing, time-series analysis
Programming	Python (pandas, NumPy, SciPy, statsmodels, arch, Streamlit, Plotly), SQL, R, MATLAB, Excel (Advanced), Jupyter Notebook, Git/GitHub
Quantitative	Financial modelling, option pricing, volatility modelling (ARCH/GARCH), credit risk (EL/PD/LGD/EAD), model validation, regression analysis, process automation, research & insight generation

EXPERIENCE

Data Analytics Intern (Extern) — Beats by Dr. Dre May – Aug 2025

- Applied Python and SQL to analyze large-scale consumer survey and market datasets, building quantitative models to surface pricing sensitivity insights and competitive positioning signals that informed product strategy decisions.
- Automated existing analytical workflows and developed dashboards to communicate complex findings clearly to executive stakeholders, improving efficiency and insight delivery.
- Collaborated cross-functionally within a high-performance team to evaluate consumer behaviour patterns across product lines, identifying opportunities to optimize pricing and drive measurable business outcomes.
- Maintained thorough documentation of analytical methodologies, ensuring reproducibility and alignment with internal governance standards.

Market Research Intern (Extern) — RevRoad Sep – Oct 2024

- Conducted in-depth capital markets research and competitive analysis to evaluate growth opportunities and investment viability for emerging companies.
- Produced structured research reports incorporating financial modelling, market risk considerations, and due diligence findings, presented to senior professionals to support transaction decisions.

Delegate Coordinator — Quantify Risk and Insurance, York University Jan – May 2024

- Organized and coordinated finance-focused academic events, demonstrating project management, teamwork, and communication skills in a client-facing capacity.
- Collected and analyzed post-event data to generate insights that improved future programming, applying a process-oriented mindset to operational challenges.

Treasurer / Financial Director — African Students' Association, York University Oct 2023 – Jul 2024

- Managed a ,000+ budget, produced transparent financial reports, and implemented process improvements that enhanced cost efficiency and governance while demonstrating accountability and integrity.
- Presented financial summaries clearly to diverse stakeholders, building written, oral, and presentation communication skills directly applicable to client-facing roles.

Finance Intern — Global Health Systems Solutions, Cameroon May – Aug 2022

- Supported financial reporting, reconciliation, and invoice tracking, demonstrating attention to detail and ability to deliver accurate outputs within structured operational processes.
- Reduced monthly reporting turnaround time through workflow standardization, reflecting a process-oriented approach to improving efficiency under competing demands.

PROJECTS

Integrated Risk App — github.com/sensor-aae/Integrated-Risk-App 2025 – Present

- Built a validation-grade market and credit risk engine in Python, structured as an importable package (risklib/) with strict separation between modelling logic, configuration, and a live Streamlit interface deployed on Render.
- Implemented four VaR/ES methodologies (Historical Simulation, Parametric, Monte Carlo with covariance shrinkage, Filtered Historical Simulation) with multi-confidence-level analysis (95%, 97.5%, 99%) and component VaR decomposition via Euler allocation.

- Extended backtesting with Christoffersen (1998) serial independence and joint conditional coverage tests ($LR_{cc} \sim \chi^2(2)$) alongside Kupiec POF, enabling detection of exception clustering invisible to frequency-only tests.
- Estimated GARCH(1,1) parameters via MLE (scipy.optimize L-BFGS-B, multiple restarts) as a data-driven upgrade to fixed-parameter volatility filtering; validated against known true parameters showing 4x lower estimation error on alpha vs. hardcoded defaults.
- Documented model assumptions, limitations, and resolution status in a structured Known Limitations section mirroring SR 11-7 / OSFI E-23 model change log standards.

Volatility Modeling with ARCH/GARCH — github.com/sensor-aae/Volatility-modeling2025

- Fitted a GARCH(1,1) model to S&P 500 daily log-returns via MLE using the arch library, demonstrating volatility clustering and high persistence ($\alpha + \beta \approx 1$) consistent with documented equity return behaviour.
- Compared MLE-estimated parameters against constant-variance and fixed-parameter baselines; MLE conditional volatility path serves as the methodological foundation for the FHS VaR module in the Integrated Risk App.

European Option Pricing — Black-Scholes — github.com/sensor-aae/European-Option-Pricing2025

- Implemented closed-form Black-Scholes pricing for European call and put options with full Greeks analysis (Delta, Gamma, Vega) and sensitivity surface visualisation across strike and spot price dimensions.
- Deployed an interactive Streamlit app for real-time parameter exploration, illustrating how option sensitivity profiles shift with moneyness and volatility.

ADDITIONAL

French (bilingual) • Active interest in global financial markets: FX, fixed income, equities, and derivatives